

Total No. of Questions : 8]

SEAT No. :

PE-2279

[Total No. of Pages : 2

[6584]-183

B.E. (Information Technology)

WIRELESS COMMUNICATION

(2019 Pattern) (Semester - VII) (414445 D) (Elective - IV)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1) a)** Explain the Principle of TDMA. What are different features of TDMA? **[9]**
- b)** What is MIMO? Explain two formats of MIMO. **[9]**

OR

- Q2) a)** What is OFDM technique? Also, explain OFDMA transmitter and receiver. **[9]**
- b)** How Code Division Multiple Access Technique is implanted while accessing a channel for multiple users? Support your theory answer with example. **[9]**
- Q3) a)** What are the different challenges in WAP? Also, write down the advantages and disadvantages of WAP. **[9]**
- b)** What is LoRaWAN? Elaborate LoRaWAN network elements. **[8]**

OR

- Q4) a)** What is NFC? What are the different characteristics of NFC? **[9]**
- b)** What is Li-Fi? Explain the working principle of Li-Fi technology. **[8]**

P.T.O.

- Q5) a)** Explain security issues and challenges in GSM. [9]
- b)** Explain in details Visible Light Communication. Also, explain its applications. [9]

OR

- Q6) a)** What is multimedia security? Explain multimedia security in 5G and 6G. [9]
- b)** Explain Bluetooth Security Architecture in detail. [9]

- Q7) a)** What is quantum Technology? Explain quantum Technology for a 5G/6G wireless network? [9]
- b)** Explain Simultaneous Transmission and Reflection (STAR) for 360° coverage in details. [8]

OR

- Q8) a)** Explain how 5G network works along-with its benefits. [9]
- b)** State and explain various applications of wireless technology. [8]
